13 Prove algebraically that the difference between any two different odd numbers is an even number.

Let the two odd numbers be $2n + 1$ and $2m + 1$,

where n and m are different integers.

1 mark
Two general odd numbers

Difference $= (2n + 1) - (2m + 1)$

$= 2n + 1 - 2m - 1$

1 mark
Subtract and factorise

$= 2n - 2m$

$= 2(n - m)$.

$n - m$ is an integer, so $2(n - m)$ is even.

1 mark
Multiple of 2 so even

(Total for Question 13 is 3 marks)